Matthew Oj

London, UK | +44 7548 238672 | edafeoj@gmail.com | linkedin.com/in/matthew-oj | github.com/Matthew-Oj | matthew-oj.github.io/

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, HTML, CSS, Bash | Familiar: C#, C++
Frameworks: FastAPI, React, Next.js, Astro, Tailwind CSS, Laravel, Streamlit, NiceGUI, pytest
Tools & Cloud: AWS (Lambda, EC2, S3, Aurora RDS, DynamoDB), GCP (Cloud Run, BigQuery, Cloud Composer, Vertex AI), Docker, Vercel, Git, GitHub Actions, CI/CD, Linux, PostgreSQL, Redis, Kafka, Snowflake, dbt, Apache Airflow, REST APIs, OAuth, JWT,

EXPERIENCE

Vafo Digital — **London, UK**
Senior Data Engineer (Backend Infrastructure) | Python, FastAPI, AWS, Docker, Snowflake — Apr 2026 – present

- Led migration of 200+ pipelines and integration services (ETL, reverse ETL, dbt, ML) to a Dockerised AWS EC2 platform with isolated staging/prod environments, an Aurora RDS metadata store and IAM-scoped access
- Migrating client analytics apps from Streamlit to a Next.js, TypeScript and Tailwind web app on Vercel, querying Snowflake via its Node.js connector
- Built a shared Python library standardising error handling and logging across 30+ services and 4 integrations (Klaviyo, DynamoDB, SFTP, S3)
- Built a Snowflake-native MCP server giving Cortex AI read-only access to Snowplow and Meta Ads data for personalised ads

Data Engineer | Python, AWS Lambda, Snowflake, dbt, Airflow, PHP/Laravel — Apr 2025 – Apr 2026

- Architected a FastAPI/Lambda (Mangum) reporting service automating scheduled delivery to 10 client accounts
- Built Python AWS Lambda functions syncing Snowflake data to HubSpot and Klaviyo APIs for 250K+ users
- Built a dbt documentation portal (HTML, CSS, JavaScript) on AWS Amplify, with GitHub Actions auto-publishing docs on merge to main
- Built 3 client-facing analytics apps (Streamlit, Snowpark) on Snowflake, querying 200M+ row event tables
- Productionised 4 ML pipelines with Data Science (dbt features, XGBoost/SHAP, Airflow, Snowflake Model Registry)

Maison Privé Concierge Limited — **Remote**
Software Engineer (Freelance) | TypeScript, Astro, Tailwind, Airtable, GCP, Vercel | Link — Apr 2026 – Jun 2026

- Built a concierge site with another developer, integrating Mailgun, Google Drive and Airtable with retries and idempotency
- Engineered an enquiry form with per-service schemas, strict validation and spam filtering blocking 95%+ of bots
- Wrote end-to-end tests per integration so partial writes and service failures stayed invisible to users

EssenceMediacom — **London, UK**
Data Engineering Executive | Python, GCP, Cloud Run, BigQuery, Airflow, GitHub Actions — Nov 2023 – Apr 2025

- Built a full-stack app (NiceGUI, Redis, BigQuery, OAuth) for client data-maturity assessments, used across 5+ WPP agencies
- Built a Python/Streamlit tool on Cloud Run generating Adverity pipeline schemas, cutting pipeline build time by 30%
- Cut Airflow DAG deploys by 40% via GitHub Actions syncing only changed DAGs (git diff + gcloud) to Composer's GCS bucket
- Built a REST API alerting service on Cloud Run monitoring Adverity failures from BigQuery, alerting teams within seconds

University of Greenwich — **London, UK**
IT Analyst | HTML, CSS, Azure, Microsoft Entra ID — Jan 2021 – Oct 2023

- Built static HTML/CSS pages for helpline scripts and common requests, cutting resolution time for frequent issues
- Resolved 20–30 support tickets a day and handled MFA and identity resets in Microsoft Entra ID (Azure AD)

PROJECTS

Finance Tracker | FastAPI, dbt, PostgreSQL, Next.js, Docker | Link

- Built a full-stack finance app: dbt models into PostgreSQL marts, a JWT-secured FastAPI API and a Next.js dashboard
- Wrote 13 pytest tests covering auth, tokens, endpoints and pagination against a mocked database; containerised with Docker Compose

Event-Driven Notification Service | Python, FastAPI, Kafka, aiokafka, PostgreSQL, Docker | Link

- Built an async notification service consuming Kafka order events, with idempotent PostgreSQL upserts for at-least-once delivery
- Added retries with a dead-letter queue for failed messages, exposing order and notification status via a FastAPI API

ACTIVITIES

WPP & Google Cloud Hackathon (3rd Place) | Python, Cloud Run, Vertex AI, BigQuery, Streamlit — Sep 2024

- Built a Cloud Run service turning live BigQuery data into Vertex AI prompts to generate ad creatives via a Streamlit UI

Global Game Jam Hackathon | ML & Game Developer | Python, OpenCV, C#, Unity — Jan 2024

- Built a Python/OpenCV laugh detector from camera input for the humour theme, and Unity/C# game objects and animations

Engineers Without Borders | Project Lead Engineer | MATLAB, SolidWorks — Sep 2020 – May 2021

- Led 5 engineers designing a solar cooker prototype with phase-change materials in SolidWorks
- Wrote a MATLAB material-selection optimiser that improved the cooker's thermal efficiency by 20%

EDUCATION

University of Greenwich — Sep 2020 – May 2023
Bachelor of Engineering, Civil Engineering, 2.1 Upper Second-Class Honours — London, UK
Dissertation (First Class): Assessing the vulnerability of catchments during flash flood events in South England using Machine Learning methods